

Toward Quantum Competitiveness in Adiabatic Quantum Computing: A Comparative Study of Variational and Factoring Algorithms

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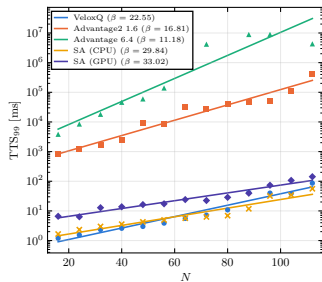
1

Local Context

- Institute of Applied and Theoretical Informatics, Polish Academy of Sciences
- Quantum Systems for Informatics group
- Work on Adiabatic Quantum Computing, simulation of many-body quantum systems

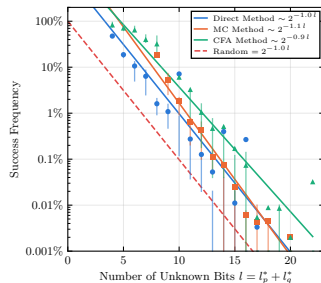


- Previous results on *exact solving* were not very promising
- Problems where a wide range of samples, not necessarily the exact solution, can be exploited



(a) Time-to-success, quantum simulation

doi:10.1103/rff3-jm5m



(b) Success rate prime factorization

arXiv:2410.14397

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Quantum Annealing

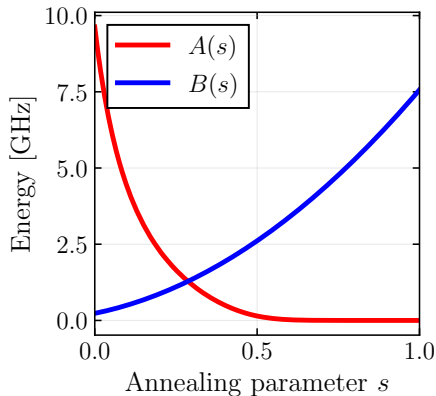
Driver Hamiltonian

$$H_D = - \sum_{i=1}^N \sigma_x^{(i)} \text{ with ground state } |+\rangle^{\otimes N}$$

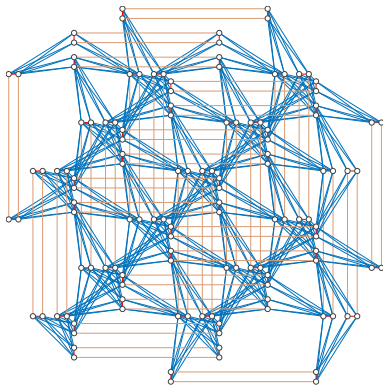
Problem Hamiltonian

$$H_P = \sum_{i=1}^N h_i \sigma_z^{(i)} + \sum_{i=1}^N \sum_{j=1}^{i-1} J_{ij} \sigma_z^{(i)} \otimes \sigma_z^{(j)}$$

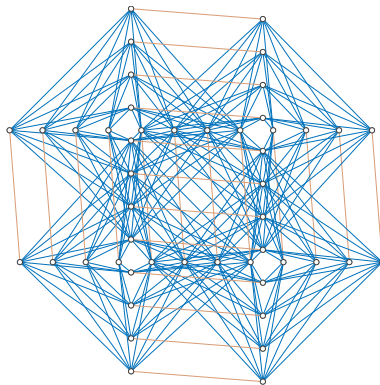
Stays in the ground state according to the adiabatic theorem [1, 2]



Annealing schedules $A(s)$, $B(s)$



(a) Pegasus (Advantage)



(b) Zephyr (Advantage2)

Circles: qubits; blue: internal couplers; orange: external couplers.
Zephyr has more couplers per qubit and longer-range links.

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Factoring Problem

Definition

Given a composite integer N , find its non-trivial factors p and q such that

$$N = p \times q$$

Methods

- Modified multiplication table method — Jiang et al., 2018 [3]
- Controlled full-adder (CFA) method — Ding et al., 2023 [4]

Based on my bachelor's thesis [5]; see also [6]

Modified Multiplication Table Method

	2^6	2^5	2^4	2^3	2^2	2^1	2^0
p					1	p_0	1
q				1	q_1	q_0	1
+					1	p_0	1
+				q_0	$p_0 q_0$	q_0	
+			q_1	$p_0 q_1$	q_1		
+		1	p_0	1			
+	-	-	-	-	-	c_0	-
+							
+			c_2	c_1			
+		c_4	c_3				
+	c_6	c_5					
+	c_7						
N	1	0	1	1	0	1	1

(a) Block size 1: 8 carry bits

	2^6	2^5	2^4	2^3	2^2	2^1	2^0
p					1	p_0	1
q				1	q_1	q_0	1
+					1	p_0	1
+				q_0	$p_0 q_0$	q_0	
+			q_1	$p_0 q_1$	q_1		
+		1	p_0	1			
+	-	-	-	-	-	-	-
+			c_1	c_0			
+							
+	c_3	c_2					
+							
N	1	0	1	1	0	1	1

(b) Block size 2: 4 carry bits

3×4 -bit multiplication table for $N = (1011011)_2 = 91$ [5, 3]

Modified Multiplication Table Method

From block size 2 (b) we obtain one equation per block:

$$\begin{aligned} 2(1 + p_0q_0 + q_1 - (4c_1 + 2c_0)) + p_0 + q_0 &= (01)_2 = 1 \\ 2(q_1 + p_0 + c_1 - (4c_3 + 2c_2)) + q_0 + p_0q_1 + 1 + c_0 &= (11)_2 = 3 \\ 2c_3 + 1 + c_2 &= (10)_2 = 2 \end{aligned}$$

Cost function

$$f(p, q, c) = \sum_k (\text{LHS}_k - \text{RHS}_k)^2, \quad \min f = 0 \iff N = p \times q$$

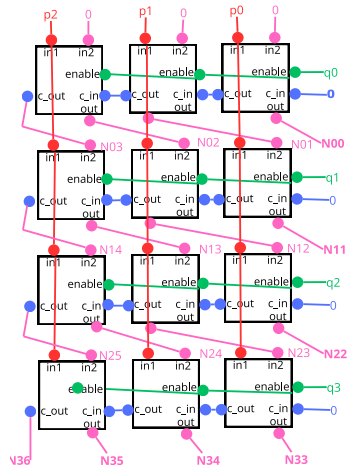
- **Reduce:** higher-order terms (e.g. $p_0q_0 \cdot p_0q_1$) to quadratic via auxiliary qubits

Source: [5, 3]

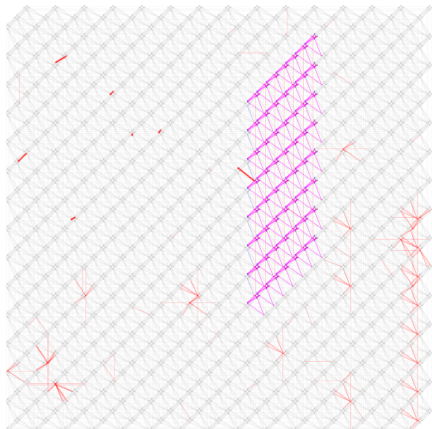
CFA gate

$$2 c_{\text{out}} + \text{out} = (\text{enable} \wedge \text{in}_1) + \text{in}_2 + c_{\text{in}}$$

		p_2	p_1	p_0	$*q_0$
	N_{03}	N_{02}	N_{01}	N_{00}	+
	p_2	p_1	p_0		$*q_1$
	N_{14}	N_{13}	N_{12}	N_{11}	+
	p_2	p_1	p_0		$*q_2$
	N_{25}	N_{24}	N_{23}	N_{22}	+
	p_2	p_1	p_0		$*q_3$
	N_{36}	N_{35}	N_{34}	N_{33}	



Implementation of a 3×4 -bit multiplier with CFAs [4, p. 5]; see [5]

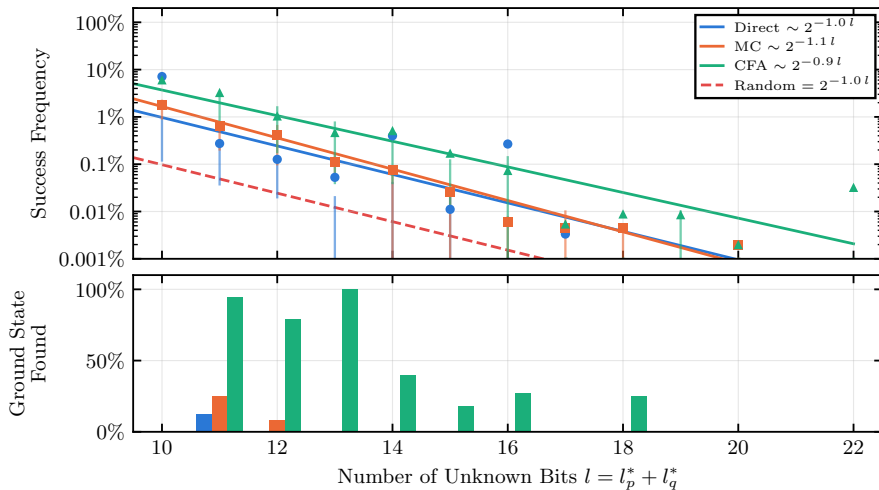


Advantage 5.4 QPU: 7×8 -bit multiplier

Source: [5, 4]

- *direct* into the hardware graph
- One CFA per unit cell

Factoring Results



Top: success frequency [6]. Bottom: runs with ground state in samples [5]

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Quantum Ising Models

TFIM Hamiltonian [7]

$$H = -J \sum_i \sigma_i^z \sigma_{i+1}^z - h \sum_i \sigma_i^x$$

- we already saw something similar earlier ;)

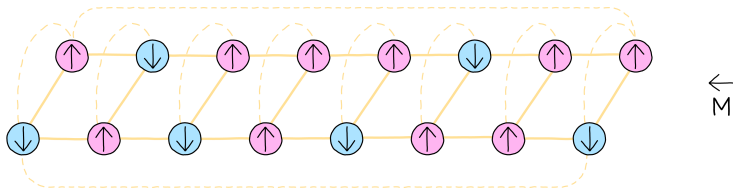


Image: PennyLane Datasets, <https://pennylane.ai/datasets/transverse-field-ising-model>

Heisenberg Hamiltonian [8]

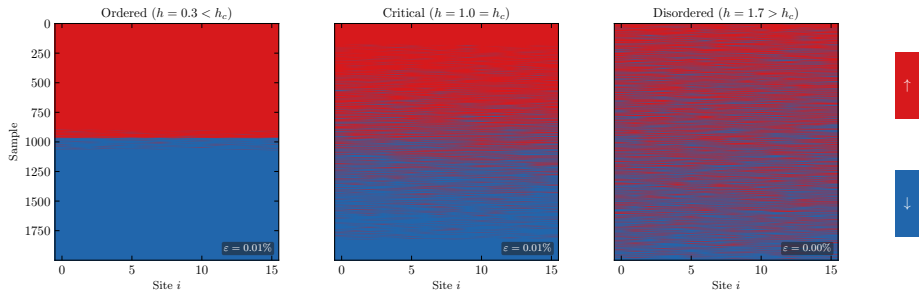
$$H = J_1 \sum_i \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+1} + J_2 \sum_i \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+2}$$

$\boldsymbol{\sigma}_i = (\sigma_i^x, \sigma_i^y, \sigma_i^z)$, so $\boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j = \sigma_i^x \sigma_j^x + \sigma_i^y \sigma_j^y + \sigma_i^z \sigma_j^z$

- main difference to Ising: spin exchange instead of flipping, next nearest neighbor coupling
- goal in both cases: finding the **ground state** (and sampling from it)

TFIM Hamiltonian [7]

$$H = -J \sum_i \sigma_i^z \sigma_{i+1}^z - h \sum_i \sigma_i^x$$



samples from a trained model with 16 spins

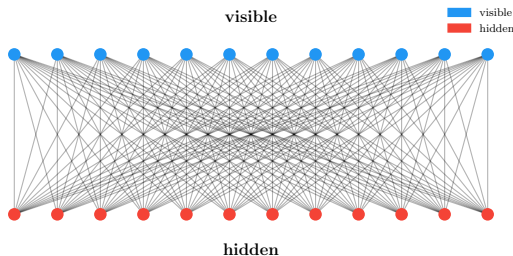
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Restricted Boltzmann Machines

- classical neural network with a layer of visible and hidden units [9, 10, 11]

RBM energy function

$$E_{\theta}(\mathbf{v}, \mathbf{h}) = \mathbf{a} \cdot \mathbf{v} - \mathbf{b} \cdot \mathbf{h} - \mathbf{h} \cdot \mathbf{W}^{\top} \cdot \mathbf{v}$$



Fully connected RBM with 12 visible and 12 hidden units

1. Sampling from the Boltzmann distribution

RBM wavefunction: hidden units traced out

$$|\Psi\rangle = \sum_{\mathbf{v}} \Psi(\mathbf{v}) |\mathbf{v}\rangle, \quad \Psi(\mathbf{v}) = \sqrt{\sum_{\mathbf{h}} e^{-E_{\theta}(\mathbf{v}, \mathbf{h})}}$$

$$\Rightarrow \Psi(\mathbf{v}) = e^{-\mathbf{a} \cdot \mathbf{v} / 2} \prod_{j=1}^{N_h} [2 \cosh(b_j + \sum_i W_{ij} v_i)]^{1/2}$$

closed form [10]

Born-rule marginal

$$\rho(\mathbf{v}) = \frac{|\Psi(\mathbf{v})|^2}{\sum_{\mathbf{v}'} |\Psi(\mathbf{v}')|^2}$$

2. Calculating the local energies

Local energy [11]

$$E_{\text{loc}}(\mathbf{v}) = \frac{\langle \mathbf{v} | H | \Psi \rangle}{\Psi(\mathbf{v})}$$

TFIM local energy [11, 10]

$$E_{\text{loc}}(\mathbf{v}) = -J \sum_{\langle i,j \rangle} v_i v_j - h \sum_i \frac{\Psi(\bar{\mathbf{v}}_i)}{\Psi(\mathbf{v})}$$

3. Calculate the log derivatives

Log-derivatives [11]

with $\phi_j = b_j + \sum_i W_{ij} v_i$

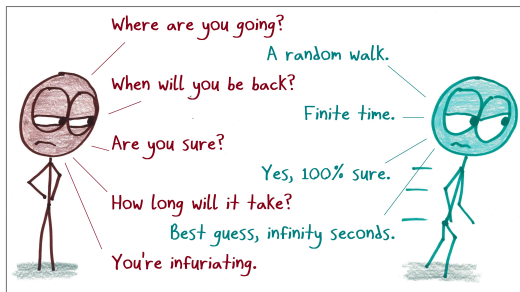
$$\frac{\partial \log \Psi(\mathbf{v})}{\partial \theta_k} = \frac{1}{2} \begin{cases} -v_i, & \theta_k = a_i \\ \tanh(\phi_j), & \theta_k = b_j \\ v_i \tanh(\phi_j), & \theta_k = W_{ij} \end{cases}$$

4. Update the parameters using GD / Stochastic Reconfiguration

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Samplers

- Classical Markov-Chain-Monte-Carlo samplers (Metropolis [12, 13]/Gibbs [14]) on GPU and FPGA
- quantum samplers: D-Wave Quantum Annealer



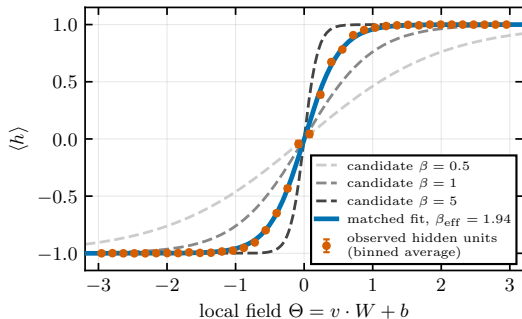
The key to any marriage is good communication.

Ben Orlin, *Math with Bad Drawings*,

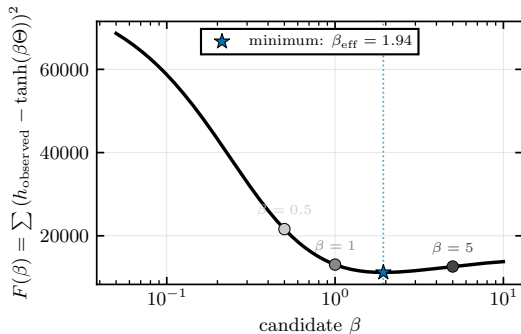
<https://mathwithbaddrawings.com/2018/10/31/im-going-on-a-random-walk/>

Temperature adjustment

- D-Wave sampler has unknown temperature
- CEM estimates the temperature from samples [15]



(a) Candidate fits vs. observed hidden units.



(b) Matching objective $F(\beta)$.

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Results

- sample matched framework
- hyperparameter-matched comparison, TFIM ($h = 0.5$), N up to 64

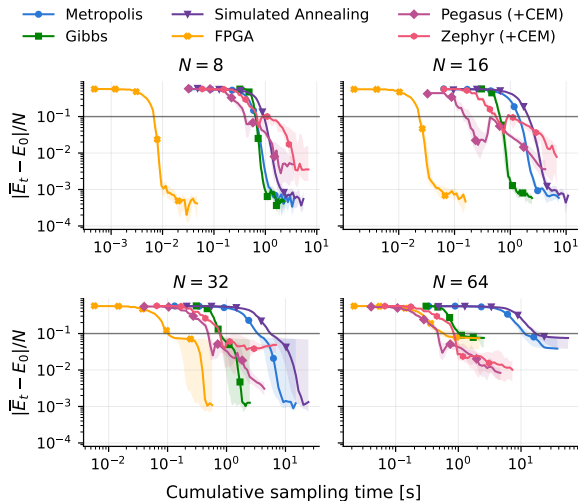
Error per spin

$$\varepsilon(t) = \frac{|\bar{E}_t - E_0|}{N}$$

$\bar{E}_t$: rolling-mean variational energy after cumulative sampling time t ; E_0 : exact ground-state energy

- median and IQR over 20 seeds per solver and size

Results: Error vs. Sampling Time



- TFIM, $h = 0.5$; horizontal line: $\varepsilon = 0.1$
- $N = 64$: Pegasus, FPGA and Zephyr cross $\varepsilon = 0.1$ within a factor ~ 2 (0.47, 0.59, 0.84 s), IQR bands overlap
- Metropolis and SA cross more than $10\times$ later (11.5, 17.1 s)

MERCI POUR VOTRE ATTENTION

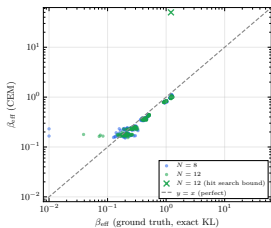


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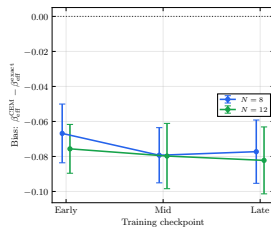
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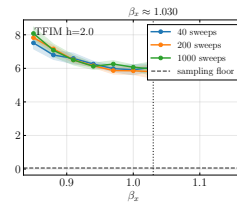
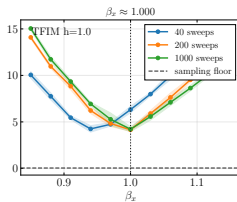
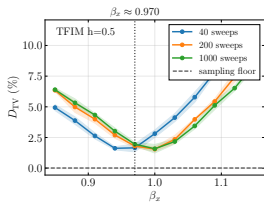
CEM & Sampling Validation



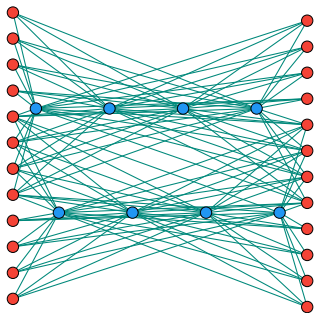
(a) CEM vs. exact ground truth



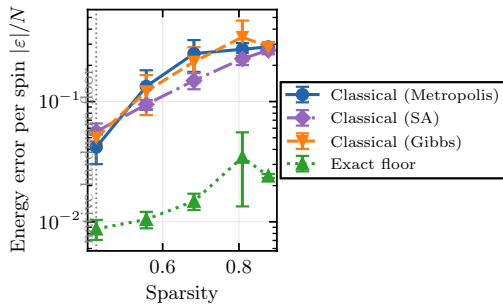
(b) Bias across training



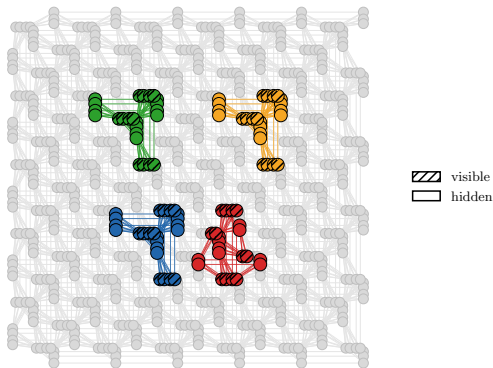
(c) D_{TV} vs. cooling sweeps



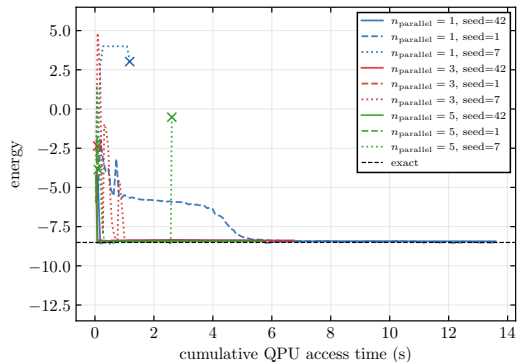
(a) Sparse RBM construction



(b) Sparsity vs. sampler error



(a) Parallel embedding layout



(b) QPU time speedup

Stoquasticity

A Hamiltonian is *stoquastic* if, in the given basis, all off-diagonal elements are real and non-positive [16].

Using $\sigma^x |\uparrow\rangle = |\downarrow\rangle$, $\sigma^y |\uparrow\rangle = i |\downarrow\rangle$, $\sigma^y |\downarrow\rangle = -i |\uparrow\rangle$, the Heisenberg exchange term acts on an antiparallel pair as

$$\left(\sigma_i^x \sigma_j^x + \sigma_i^y \sigma_j^y\right) |\uparrow\downarrow\rangle = |\downarrow\uparrow\rangle + (i)(-i) |\downarrow\uparrow\rangle = 2 |\downarrow\uparrow\rangle$$

Consequently, every antiparallel bond contributes

$$\langle\downarrow\uparrow| H_{ij} |\uparrow\downarrow\rangle = 2J > 0 \implies \textbf{not stoquastic}$$

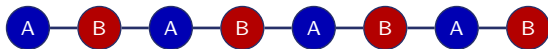
Marshall Sign Rule

- Bipartite lattice $A \cup B$: every nearest-neighbour (J_1) bond connects a site in A to a site in B
- Unitary transformation, Pauli convention [17]:

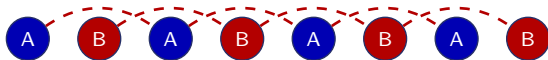
$$U = \prod_{i \in A} \sigma_i^z$$

- Ground-state wavefunction after transformation (only positive amplitudes):

$$U|\psi_0\rangle = \sum_{\sigma} |c_{\sigma}| |\sigma\rangle$$



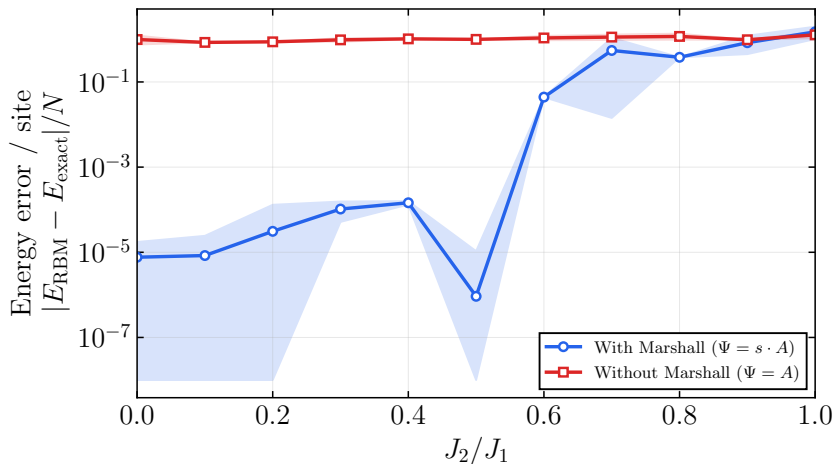
(a) J_1 : always $A-B$



(b) J_2 : same sublattice $\rightarrow$ breaks bipartiteness

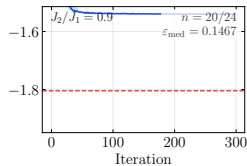
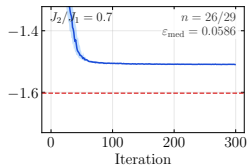
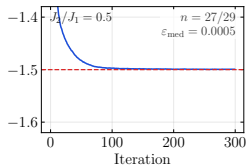
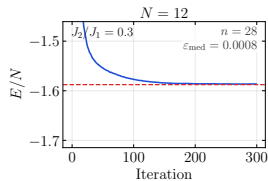
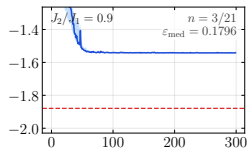
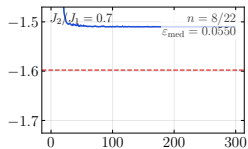
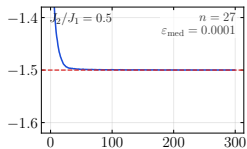
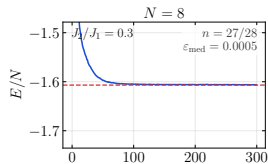
Marshall correction survives moderate frustration [18, 19]

Marshall Sign Correction



Energy error per site vs. J_2/J_1 , with/without correction

RBM Convergence Curves



E/N vs. SR iteration, frustrated Heisenberg chain